

GTSC2143 Machine Learning for Business

Tutorial 10

Please write down your answers in this document and submit it at iSpace by the end of this tutorial.

Activity 1. Data Loading and Processing

1. Load the Dataset

```
import pandas as pd

# Load the Mall Customers Data

data

=pd.read_csv("https://gist.githubusercontent.com/pravalliyaram/5c05f43d2351249927b8a3f3cc3e5ecf/raw/8bd6144a87988213693754baaa13fb204933282d/Mall_Customers.csv",encoding='iso8859-1')
```

2. Explore the dataset:

- Check the data shape
- Display the first 3 rows
- Summarize columns and data types

3. Data processing

- Encode categorical variables such as Gender into numeric form (e.g. 0 for Male, 1 for Female)
- Select only numerical features relevant to clustering
- Apply feature scaling (e.g. `StandardScaler`) to ensure all variables contribute equally to the distance metric
- Remove irrelevant columns such as `CustomerID` that should not affect clustering

Activity 2. Distance Metrics and Similarity

1. Select two customers and display their information:

- Customer ID = 1
- Customer ID = 200

2. Compute manually or using Python the following distances:

- Euclidean
- Manhattan
- Chebyshev
- Cosine similarity

3. Which distance metric best represents customer similarity in this dataset? Why?

Activity 3. K-Means and Elbow Method

1. Apply ordinary K-Means Clustering

- Apply the ordinary K-Means method with `n_clusters=3`, `random_state=42`
- Color points by cluster label by Annual Income and Spending Score

2. Apply Elbow method

- Apply the Elbow Method using $k = 1 - 10$
- Plot WCSS vs k , identify the elbow point, and justify your chosen k
- Apply the ordinary K-Means method with `n_clusters=k` in b), `random_state=42`
- Color points by cluster label by Annual Income and Spending Score

- e) What's the difference between these two clustering results (interpret from the plots, if any)?
3. Apply K-Means ++ and K-Medoids
- a) Apply K-Means ++ and K-Medoids
 - b) Compare cluster centroids with K-Means using plots
 - c) Discuss robustness to outliers

Activity 4. Other Clustering Methods

1. Apply Density-Based Clustering (DBSCAN)
 - a) Apply the DBSCAN using `eps=0.6`, `min_samples=5`
 - b) Identify core, border, and noise points visually
 - c) Why DBSCAN cannot specify `random_state`?
2. Apply Gaussian Mixture Model (GMM)
 - a) Apply the GMM using `n_components=5`, `random_state=42`
 - b) Plot the GMM clusters by Annual Income and Spending Score

Activity 5. Clustering Evaluation

1. Compute the Silhouette Coefficient for all the algorithms
 - a) K-Means
 - b) K-Means++
 - c) K-Medoids
 - d) DBSCAN
 - e) GMM
2. Which model produces the highest silhouette score?
3. Why does the clustering performance in the Mall Customers dataset appear to be not so good?
Briefly explain possible reasons and how you could improve the clustering results.